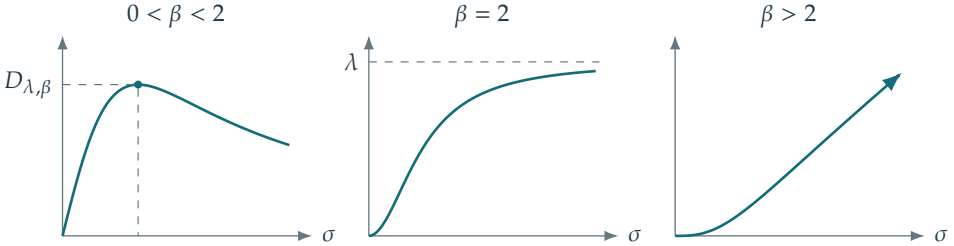
$$b_{\lambda,\beta}(\sigma) = \frac{\lambda \sigma^{\beta}}{\lambda + \sigma^{2}}, \qquad D_{\lambda,\beta} = \sup_{\sigma \geq 0} b_{\lambda,\beta}(\sigma)$$



$$\sigma_{*} = \sqrt{\frac{\beta\lambda}{2-\beta}} \qquad D_{\lambda,2} = \lambda \qquad D_{\lambda,\beta} = +\infty$$

Here $\lambda > 0$ and the supremum ranges over all $\sigma \geq 0$.
On a fixed bounded operator, only $0 \leq \sigma \leq \|\Phi\|$ is relevant.